

Ritwik Kumar

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EDUCATION

SRM Institute of Science and Technology

2023 – 2027

B.Tech, Computer Science Engineering with specialization in Big Data Analytics

Chennai, India

CGPA: 9.16 / 10 (Current)

Coursework: Data Structures, Algorithms, OOPs, Data Science, DBMS, AI, ML · Certifications: Web Dev 3.0 (Udemy) · ML Specialization (DeepLearning.AI)

PROJECTS

Formula-1 Analysis & Predictions Dashboard

Jul 2025 – Nov 2025

Python, Pandas, Scikit-Learn, Lasso Regression, Clustering, Streamlit, Jupyter

- Ingested and cleaned 45+ years of F1 race data (lap times, pit stops, tyre strategies, driver stats) across multiple seasons using automated pandas ETL pipelines with modular feature engineering.
- Built Lasso Regression predictive models achieving $R^2 > 0.85$ to forecast championship standings; applied temporal analysis and k-means clustering to identify performance patterns across circuit types and conditions.
- Deployed a full interactive Streamlit web app with real-time filtering, multi-season comparison views, and driver-vs-team benchmarking dashboards for data-driven race strategy insights.

Fuel-Per-Kilometer Analysis

Dec 2024 – May 2025

Python, Pandas, Matplotlib, Scikit-Learn, Geospatial Analysis, Streamlit

fuelsense.streamlit.app

- Processed and analyzed fuel-consumption-per-kilometer data across diverse vehicle categories and geographic regions; applied regional clustering and geospatial mapping to uncover consumption pattern differences.
- Engineered modular preprocessing and feature extraction pipelines; implemented regression models to predict fuel efficiency and quantify the impact of vehicle type, load, and terrain on consumption.
- Deployed an interactive Streamlit dashboard with dynamic charts, geospatial visualizations, and region-wise comparison tools, delivering actionable insights for fuel economy optimization and sustainable decision-making.

LEADERSHIP & ACTIVITIES

Cherry+ Network — Corporate Associate

Oct 2024 – Present

Chennai, India

- Secured 10+ sponsors for major tech events through end-to-end stakeholder outreach, proposal writing, and budget negotiation.
- Led full-cycle organization of ForceFest (Cybersecurity Workshop) and the Nariyukthi Hackathon — managing vendor coordination, logistics, and on-ground execution without cost overruns.

AARUUSH — Championships Volunteer

Aug 2023 – Oct 2023

Chennai, India

- Contributed to planning for flagship events TurboDrift and QuadCopter; coordinated stage setup, team logistics, and live event flow ensuring smooth execution for 500+ participants.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, JavaScript, HTML/CSS, SQL, Bash
Databases	MySQL, PostgreSQL, MongoDB, Snowflake
ML & Data	Regression & Classification, Clustering, Time-Series Forecasting, Ensemble Methods (XGBoost, LightGBM, Random Forest), Deep Learning (CNNs, RNNs, LSTMs, Transformers), NLP, Bayesian Modeling, Hypothesis Testing, Spark / Hadoop
Frameworks	TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy, Streamlit, FastAPI, Kafka, Docker, Git
Other	ETL Pipelines, MLOps & Model Deployment, REST APIs, OOP, Technical Documentation